

# Landscape-scale bacterial biogeography across the Rub' al-Khali

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## ABSTRACT

The Rub' al-Khali (Empty Quarter) is the world's largest continuous sand desert, and its soil microbiome has not been surveyed across its vast landscape or over repeated visits. We surveyed microbial communities at 60 sites along a west to east transect through the Empty Quarter in Saudi Arabia during 5 campaigns over 31 months. At each site, we sampled surface, shallow-subsurface, and root-adjacent soil, and obtained 1,237 16S rRNA profiles together with climate, soil pH, and soil chemical composition data. Microbial community composition changed across the landscape, where dissimilarity increased with geographic distance in all sampled compartments. Replacement of taxa between sites accounted for most of the dissimilarity in taxon presence and absence. Within sites, root-adjacent soil had the lowest alpha diversity and harbored communities distinct from bulk soil. Climate, soil pH, and soil chemical composition changed along the transect along with community composition. Exploratory models associated higher richness with rainfall in the days before sample collection. Predicted metabolic pathways followed the same geographic and compartment patterns. This survey and its linked data resource provide a baseline for the Empty Quarter soil microbiome and a framework for targeted environmental measurements and repeated surveys in the region.

**Keywords:** Rub' al-Khali, Empty Quarter, hyperarid soil microbiome, bacterial biogeography, landscape-scale survey, spatiotemporal, repeated sampling, 16S rRNA amplicon survey

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## INTRODUCTION

Drylands are among the most extensive terrestrial biomes [1], and aridity is projected to increase across much of them [2]. Where plants are sparse and rainfall is rare, soil microbial communities carry much of the remaining biological activity [3]. Soil microbes influence soil fertility and crop production, and drive environmental processes such as carbon and nitrogen fixation, organic matter decomposition, and greenhouse gas emissions [4, 5]. Bacteria isolated from dryland soils have been used to improve plant tolerance to abiotic stress [6] and to stabilize and restore degraded soils [1]. Therefore, characterizing the composition and diversity of dryland soil microbiomes is relevant to agriculture, conservation, and land management.

The Rub' al-Khali (Arabic الربع الخالي; in English, the Empty Quarter) is the world's largest continuous sand desert. It covers much of the southern Arabian Peninsula, mainly within Saudi Arabia, and is one of the least explored drylands in the world [7]. It is a hyperarid desert where rainfall is rare and temperatures approach 50°C, thus considered one of the most hostile environments on Earth [8]. It is characterized by sand dune and saline pans referred to as *sabkha*, and sparse shrubs and fauna [9]. Ancient lakebeds are found throughout the desert, although it has no present link to rivers or other large water bodies [7]. Due to its unique terrain and climate, as well as its isolation from human influences, the Empty Quarter can serve as a model of desertification. Desertification is characterized by a decline in soil fertility and reduction in grass vegetation [2], both of which are characteristic of the Empty Quarter, and understanding microbial community adaptations to this harsh habitat could play an important role in remediation efforts.

Desert soils have been studied at different scales. At large spatial scales, community composition differs between regions and along aridity gradients. In the Mojave desert, biological soil crust communities from central and southern sites had distinct compositions [10]. In the Atacama and Gobi deserts, the colonization and diversity of hypolithic communities declined as aridity increased [11, 12]. Community composition also varied between habitats within a landscape, such as in the Namib desert, where a comparison of sand dune zones showed that each zone harbored a different bacterial community [13]. At a smaller scale, some studies have shown that communities differed between surface and deeper soils. Surface and subsurface communities were distinct in Mojave soil crusts [10], and microbial biomass in an Atacama playa decreased with depth [14]. Also within sites, microbial communities differed between bulk soil and plant-associated soil, where the rhizosphere held more diverse communities than surrounding bulk soil in the Namib desert [15].

These spatial patterns have also been studied in relation to soil properties and environmental conditions. The distinct microbial communities in the Namib sand dunes varied with habitat stability, soil texture, and mineral and nutrient content [13]. Decline in biomass with the depth gradient in the Atacama was linked to an increase in soluble salts [14]. The Namib rhizosphere was found to be cooler, more humid, and richer in nutrients than bulk soil [15].

Desert soils have also been studied over time. Rainfall in deserts is rare, usually in short pulses, and the pulse-reserve model proposes that each pulse triggers a brief period of biological activity before the soil dries again [16, 17]. In the Atacama desert, heavy rainfall re-distributed salts in the soil, and shifted community composition to be more similar to less arid sites [18]. Temperature has also been found to affect communities over time, where hypolithic communities in the Gobi desert were found to significantly differ from those sampled after cold periods [12].

Studies that combine repeated sampling and large spatial scales in desert soils remain sparse [19], and the Arabian Peninsula remains underrepresented in global studies [2, 20]. In the Arabian Peninsula, studies have described microbial communities along elevation gradients [21], at single locations [22], in plant-associated soil [23, 6, 24], and in an inland sabkha of the Empty Quarter in the United Arab

Emirates [25]. The Empty Quarter desert has not yet been surveyed across its landscape or over repeated visits. Here, we surveyed the microbial communities of the Empty Quarter in Saudi Arabia in 60 sites along a west to east transect during 5 campaigns over 31 months. At each site, we sampled surface, shallow-subsurface, and root-adjacent soil, and recorded climate conditions and environmental features along with 16S rRNA sequencing of the microbial communities. We also obtained soil pH and chemical composition of the samples across compartments and campaigns.

We found that bacterial community composition changed across the landscape. Dissimilarity increased with geographic distance in all three compartments, and the replacement of taxa, or turnover, accounted for most of the dissimilarity. Within sites, root-adjacent soil harbored distinct communities from bulk soil, and had lower alpha diversity. Climate, soil pH, and soil chemical composition changes along the west to east transect along with community composition. Exploratory models associated richness with rainfall in the days before collection. Predicted metabolic pathways followed the same geographic patterns. We also found that PMA-accessible DNA contributed to measured richness in an exploratory experiment. The survey and its linked data resource [26] provide a framework for targeted environmental measurements and repeated surveys. All analysis inputs and code are available in the ecology reproducibility repository.

## RESULTS

### **Pseudomonadota, Bacillota and Actinomycetota dominate Empty Quarter soils**

We surveyed 60 sites along a west to east transect through the Empty Quarter in Saudi Arabia during 5 campaigns over 31 months (Figure 1a). At each site, we sampled bulk surface soil, shallow-subsurface soil and root-adjacent soil, collecting three field replicates per compartment. We sequenced the V3–V4 region of the bacterial 16S rRNA gene and resolved the reads into amplicon sequence variants (ASVs). This yielded 1,237 samples, 1,227 of them from the repeatedly sampled sites 1–60 (Figure 1b). Trip 2 covered only 8 sites due to extreme heat.

We assessed contamination at different study stages, and found that candidate contaminant ASVs comprised 2.19% of pooled reads in 217 linked T5 profiles; before/after Shannon diversity and raw-richness rankings were strongly correlated ( $p = 0.995$  and  $0.99995$ , respectively). Samples in adjacent wells during preparation were no more similar than matched samples in non-adjacent wells (one-sided  $p = 0.87$ ). Mock community controls varied in their recovery of expected genera. We therefore retained all ASVs in downstream analyses (Supplementary Section S2).

We first describe the overall microbial community composition of the 1,227 samples. Pseudomonadota was the most abundant phylum (26.9%), followed by Bacillota (22.5%) and Actinomycetota (18.7%). Acidobacteriota and Verrucomicrobiota, which are among the most abundant phyla in global soil surveys [5, 27], together accounted for 3.3%. At the genus level, *Domibacillus* (6.2%) was most abundant in every compartment, followed by *Massilia* (3.5%), *Bacillus* (3.0%), *Microvirga* (2.4%) and *Flavisolibacter* (2.0%). ASVs without a genus assignment made up 35.6% of reads per sample on average (Supplementary section S4.1).

### **Microbial communities change across the landscape**

To test whether community composition changed across the landscape, we modelled the composition of each site, based on the 200 most abundant genera, as a function of its position along the transect, allowing for either a steady west to east change or a change that peaks or troughs mid-transect. Transect position explained 40.1% of the variation in composition among the 60 sites (95% CI, 32.4–47.7%), compared with 25.4% for a steady west–east trend alone. The genus-level result persisted when we changed the number of genera included or omitted any one campaign; an ASV-resolution sensitivity analysis also showed a transect pattern (Supplementary Section S4.4.2). However, the model residuals showed significant spatial autocorrelation (Moran's  $I = 0.065$ ,  $p = 0.001$ ), so sites could not be treated as independent, which inflates the significance of permutation tests. When we accounted for this spatial autocorrelation, the trend remained significant if autocorrelation was assumed to be short-range, but support depended on the covariance model at ranges of 50 km or more (Supplementary Section S4.4.3). We therefore report the variance explained as a descriptive measure of how strongly composition aligns with the transect, rather than evidence for a statistically supported west to east gradient.

Microbial communities became less similar as geographic distance increased. Aitchison dissimilarity increased with distance in all three compartments (Table 1, Figure 1c), and decay rates differed among

compartments ( $p = 0.0051$ ). Communities changed faster with distance in shallow-subsurface than in root-adjacent soil ( $p_{\text{adj}} = 0.004$ ), whereas surface and root-adjacent soil did not differ ( $p_{\text{adj}} = 0.173$ ).

To determine whether differences between sites reflected replacement of genera or poorer sites containing subsets of the genera at richer sites, we partitioned Sørensen dissimilarity into turnover and nestedness components [28]. Turnover accounted for most of the dissimilarity between sites in every compartment (69–75 %; Table 1), indicating that this transect of the Empty Quarter contains distinct communities rather than progressively poorer subsets of a single community. Both components increased with geographic distance in surface and shallow-subsurface soil (all  $p = 0.0001$ ). In root-adjacent soil, turnover increased faster than nestedness (Supplementary Section S4.5). Root-adjacent soil also had the largest pool of genera shared across sites and the most genera per site (Table 1). The three compartment cores shared 59 genera (Figure 2).

**Table 1.** Distance decay, partitioning of Sørensen dissimilarity and core genera by compartment.

Aitchison slope: increase in Aitchison dissimilarity per 100 km. Sørensen, turnover and nestedness: mean pairwise dissimilarity between sites after rarefaction, with the increase per 100 km in parentheses. Core genera: genera present at  $\geq 90\%$  of sites after 20 rarefaction draws (detection in at least half of draws). Genera per site is the median across 60 sites. All common-depth calculations use 12,865 reads. Aitchison intervals use the delete-one-site jackknife.

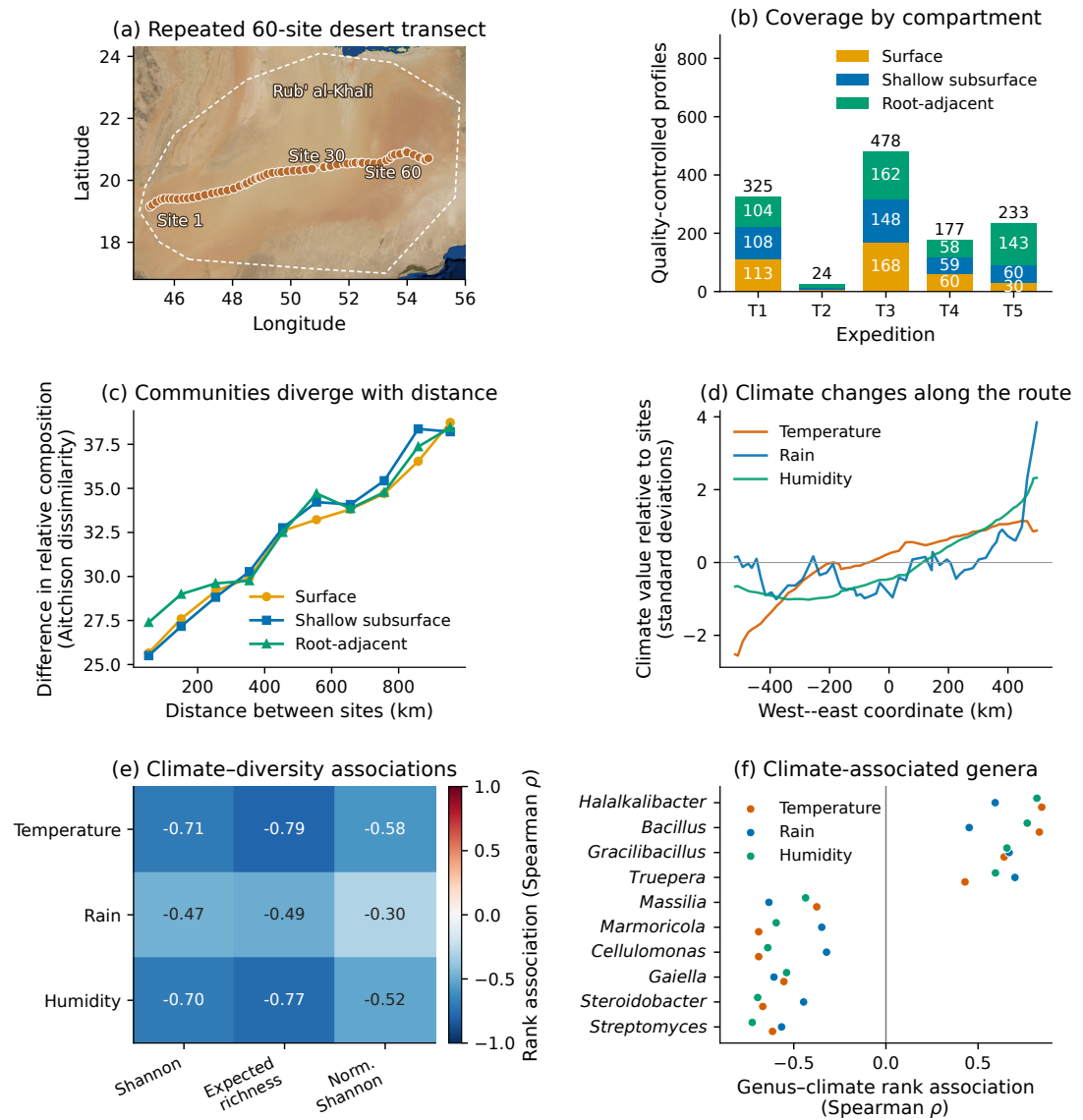
| Compartment        | Aitchison slope [95 % CI] | Sørensen | Turnover       | Nestedness     | Core genera | Genera per site |
|--------------------|---------------------------|----------|----------------|----------------|-------------|-----------------|
| Surface            | 1.35 [0.90, 1.80]         | 0.401    | 0.286 (0.0075) | 0.115 (0.0112) | 82          | 274             |
| Shallow-subsurface | 1.51 [1.09, 1.93]         | 0.406    | 0.280 (0.0096) | 0.126 (0.0164) | 69          | 284             |
| Root-adjacent      | 1.18 [0.70, 1.66]         | 0.395    | 0.295 (0.0117) | 0.100 (0.0054) | 85          | 299             |

To identify the genera associated with this gradient, we correlated the relative abundance of each of the 200 genera with transect position using CLR profiles. 124 genera were significantly associated with the transect ( $q < 0.05$ ), where 77 decreased and 47 increased from west to east (Supplementary Table S6). Genera that decreased were mainly Actinomycetota (35 %), including *Cellulomonas* ( $\rho = -0.79$ ) and *Marmoricola* ( $\rho = -0.70$ ), whereas genera that increased were mainly Bacillota (36 %). *Bacillus* increased from 1.6 to 6.2 % mean relative abundance between the western and eastern thirds ( $\rho = 0.77$ ) and became the most abundant genus in the east, together with genera typical of saline and alkaline soils, including *Halalkalibacter* ( $\rho = 0.85$ ), *Aquibacillus* ( $\rho = 0.72$ ) and *Pseudalkalibacillus* ( $\rho = 0.74$ ). The eastern end of the transect was also the warmest, most humid and most alkaline (Spearman  $\rho$  between site means and transect position: 0.98 for temperature, 0.92 for humidity and 0.82 for pH; Supplementary Table S7). The taxonomic change therefore coincides with environmental gradients that cannot be separated from geographic position in this experimental design.

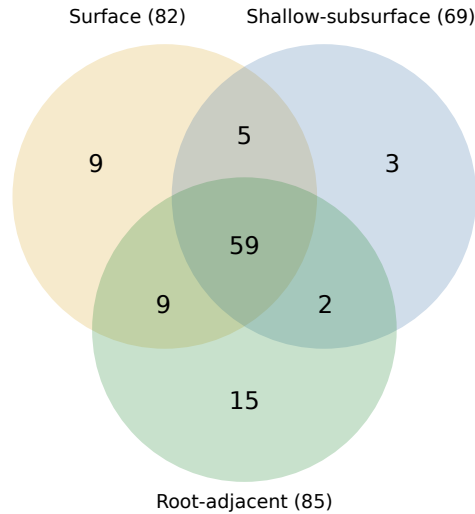
We next asked how landform contributed to the geographic pattern, since saline pans and oases are unevenly distributed along the transect. Field observations assign a landform to each site: 44 sand dunes, 9 saline pans (3 western, 1 central, 5 eastern), 3 eastern oases, and one gravel plain, dune slack, aeolian lake and oil-spill site each (Supplementary Section S4.6.3). Saline pans differed compositionally from dunes (pseudo- $F = 4.56$ ,  $p = 0.0013$ ; 16 genera at  $q < 0.05$ ), but the difference was much smaller after accounting for transect position (pseudo- $F = 1.78$ ,  $p = 0.032$ ), and no genera remained differentially abundant. Sand dunes and saline pans did not differ in alpha diversity ( $p \geq 0.28$ ), whereas oases had the lowest Shannon diversity (mean 4.25, compared with 5.77 at sand dune sites). When restricted to dune sites, the transect model still explained 34.2 % of compositional variation (40.1 % across all sites), and 70 of the 71 genera significantly associated with the transect changed in the same direction as in the full dataset (Supplementary Section S4.6.3).

### Compartment differences in composition and diversity within sites

As composition varies across the transect, we next tested whether communities differed between compartments at the same site and campaign. We used the 170 site  $\times$  campaign combinations with all three compartments, centred each combined sample profile on its mean, and averaged within site. Across the 170 complete site  $\times$  campaign blocks, the pooled test detected compartment differences (pseudo- $F = 17.28$ ,  $p = 0.001$ ), and every pair of compartments differed in composition ( $q = 0.001$ , Table 2). To assess whether these differences were consistent across sites, we calculated a standardized displacement, which approaches 1 when compartments differ in the same direction at every site, and 0 when



**Figure 1.** Survey design, geographic differences and climate associations. (a) The 60 core sites along the west-east route on a NASA Blue Marble satellite image (July 2004); the dashed outline marks the Empty Quarter boundary for orientation only. (b) The 1,237 quality-controlled profiles by campaign and compartment. Trip 2 reached 8 sites. (c) The difference in relative community composition, measured as mean Aitchison dissimilarity, as distance between sites increases. The statistical test kept all comparisons involving the same site together. (d) Each site's 49-month climate mean, expressed as standard deviations from the mean across all sites. (e) Rank associations between climate and diversity. Values near  $-1$  mean that diversity consistently decreases as the climate variable increases; values near zero mean no consistent order. (f) The strongest supported genus associations. Tests in (e,f) were corrected for the number of comparisons.



**Figure 2.** Overlap of core genera among soil compartments. A genus belongs to a compartment's core when it occurs at  $\geq 90\%$  of the 60 sites after pooling counts within site and compartment across campaigns and rarefying 20 times to 12,865 reads; detection requires presence in at least half of the draws. Numbers give mutually exclusive intersections; circle areas are schematic.

site-level differences point in opposing directions. Displacement was largest between root-adjacent and shallow-subsurface soil (0.48), followed by root-adjacent and surface soil (0.43) and shallow-subsurface and surface soil (0.33, Figure 3a). These differences remained significant with the same directions when we changed the number of genera tested, the treatment of zero counts, or omitted any one campaign (Supplementary Section S4.4.5).

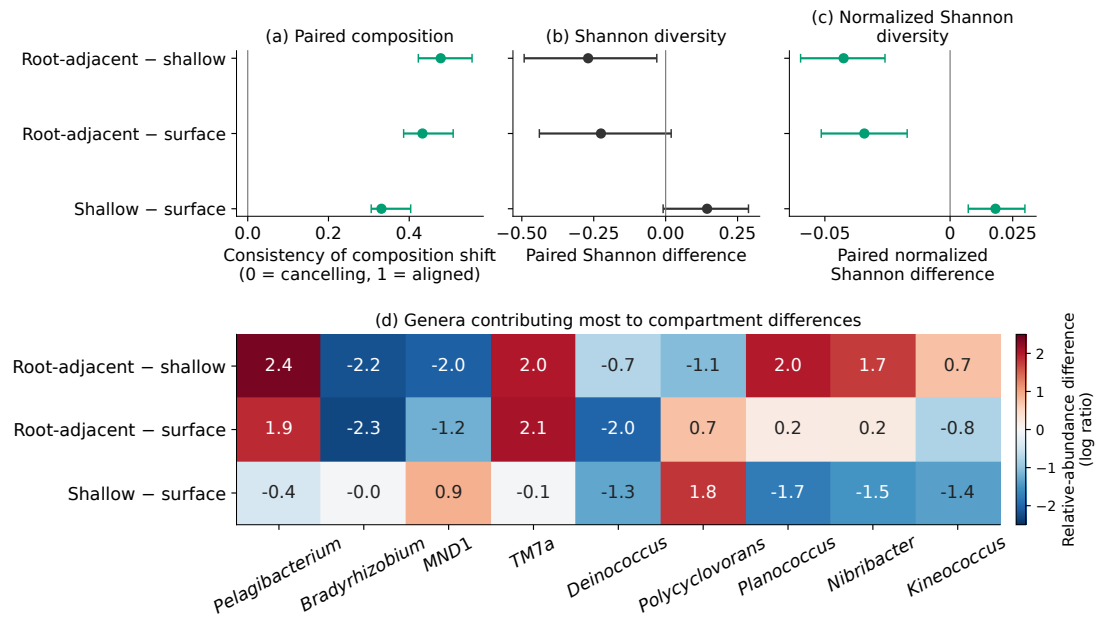
Compartments also differed in variation among site profiles within each compartment, which we tested by multivariate dispersion. Dispersion was lowest in surface soil and highest in root-adjacent soil (Table 2). Root-adjacent and shallow-subsurface soil were both more variable than surface soil ( $q = 0.003$  and  $0.015$ ), but did not differ significantly from each other ( $q = 0.061$ ). Because PERMANOVA is sensitive to differences in dispersion as well as in mean composition [29], we based conclusions about composition on the paired location tests, which compare compartments within each site and assume sign symmetry of the site-level difference vectors under the null.

Alpha diversity was lowest in root-adjacent soil (Figure 3b,c; Table 2). Compared with both bulk soil compartments, root-adjacent soil had lower rarefied richness (173 fewer ASVs than surface soil and 266 fewer than shallow-subsurface soil) and lower normalized Shannon diversity. Shannon diversity was lower in root-adjacent than in shallow subsurface soil (mean difference  $-0.27$ , 95 % CI  $-0.49$  to  $-0.03$ ). Shallow-subsurface soil had higher normalized Shannon diversity than surface soil, but they did not differ in rarefied richness. An alternative comparison that averaged each compartment across campaigns gave the same directions but weaker richness differences, and adjusting Shannon contrasts for sequencing depth and DNA extraction kit, and normalized Shannon contrasts for sequencing depth, did not change the direction of any contrast (Supplementary Sections S4.2.2 and S4.2.3).

We next identified the genera associated with these compositional differences. Because read abundances are relative, we compared each genus with the geometric mean abundance across the selected 200 genera on a logarithmic scale. Testing the 200 genera from the transect model across the three compartment contrasts (600 tests) supported 295 differences after correction (Supplementary Table S14). Surface soil was enriched in pigmented and desiccation-tolerant genera, including *Deinococcus*, *Rubrobacter*, *Kineococcus*, and *Kocuria*. Root-adjacent soil was enriched in plant-associated genera, including *Neorhizobium*, *Devosia*, *Pseudomonas*, and *Brevundimonas*. Interestingly, the root nodule-associated genus *Bradyrhizobium* was more abundant in bulk soil compartments. Shallow-subsurface soil was enriched in the nitrifying *Nitrospira* and the Nitrosomonadaceae lineage MND1, and in *Gaiella*.

**Table 2.** Pairwise differences between compartments within 60 sites. Diversity contrasts first average within matched site–campaign blocks and then across campaigns within site, using the common cohort with all three diversity measures. Richness is expected ASV richness at 25,000 reads. Displacement and dispersion use paired compositional profiles. \* $q < 0.05$  (Benjamini–Hochberg within each measure): diversity uses Wilcoxon signed-rank tests; displacement and dispersion use paired sign-flip tests.

| Contrast                           | Displacement | Dispersion | Shannon | Richness | Normalized Shannon |
|------------------------------------|--------------|------------|---------|----------|--------------------|
| Shallow-subsurface – surface       | 0.33*        | 1.38*      | 0.14*   | 110      | 0.018*             |
| Root-adjacent – surface            | 0.43*        | 2.39*      | −0.22*  | −173*    | −0.034*            |
| Root-adjacent – shallow-subsurface | 0.48*        | 1.01       | −0.27*  | −266*    | −0.043*            |



**Figure 3.** Compartment differences within sites. (a) Consistency of the shift in relative taxon abundance across sites. Values approach one when sites shift in the same direction and zero when shifts cancel. (b,c) Campaign-matched differences in Shannon diversity and normalized Shannon diversity, using profiles with all three diversity endpoints. We formed differences within campaign and site before averaging within each site. Points show means and bars show 95 % site-bootstrap intervals. (d) Genera with the largest contributions to the three composition contrasts. Positive values indicate higher CLR abundance in the first named compartment.

### **Climate covaries with the geographic pattern**

Air temperature during sampling ranged from 20.0 to 49.4 °C and relative humidity from 9.5 to 62.5 %. These single-visit measurements were not associated with alpha diversity after accounting for campaign and transect position (all  $q \geq 0.319$ ), which is expected as they capture transient weather rather than climate conditions under which communities develop (Supplementary Section S4.2.4). We therefore used long-term climate summaries averaged over 49 months for each site, where long-term mean air temperatures ranged from 26.4 to 30.3 °C, monthly rainfall from 1.3 to 7.5 mm, and relative humidity from 24.3 to 35.4 % (Figure 1d).

Sites with higher long term temperature, rainfall, and humidity had lower alpha diversity. All correlations between the climate variables and the three alpha diversity measures were negative and significant ( $q \leq 0.021$ ; Figure 1e), and were strongest for temperature and humidity. Climate variables were also associated with the relative abundance of more than half of the 200 genera tested (111–112 genera per variable; Table 3; Figure 1f). Genera most positively associated with temperature included *Halalkalibacter*, *Bacillus* and *Ammoniphilus* ( $\rho = 0.75$ – $0.85$ ), and those most negatively associated included *Cellulomonas*, *Marmoricola* and *Steroidobacter* ( $\rho = -0.67$  to  $-0.69$ ), largely the same genera that changed along the transect. Because temperature and humidity were themselves strongly correlated with transect position ( $\rho = 0.98$  and  $0.92$ ), these associations describe a shared climatic and geographic gradient rather than independent effects of climate.

### **Soil pH tracks the west-to-east change in community composition**

Soil pH was alkaline (mean 7.99) and increased from west to east. Transect position explained 84.3 % of the variation in pH after accounting for campaign and compartment ( $p = 0.001$ ). To quantify its association with the geographic pattern in composition, we partitioned variation in the adjusted mean composition of the 60 sites using the 560 campaign  $\times$  site  $\times$  compartment groups with pH measurements. With a common variance denominator, 1.38 % was uniquely associated with pH, 13.56 % uniquely with transect position, and 23.86 % shared between them; 61.20 % remained unexplained. The shared component describes the covariance of pH and geography with community composition. Within sites, the conditional association with pH was small (partial RDA, 0.28 % of variation,  $p = 0.125$ ; Supplementary Section S4.6.1).

Of the 200 genera tested, 117 were correlated with mean site pH ( $q < 0.05$ ), 104 of which were also associated with the transect. Alkaliphilic and halophilic genera, including *Polygonibacillus*, *Sediminibacillus*, *Halalkalibacter* and *Halomonas*, increased with pH ( $\rho = 0.76$ – $0.81$ ), whereas *Pirellula*, *Gemmatimonas*, *Gaiella* and *Streptomyces* decreased ( $\rho = -0.64$  to  $-0.67$ ; Supplementary Section S4.6.1).

### **Soil chemical composition is a weak correlate of community composition**

X-ray fluorescence (XRF) measured 11 elements in at least 75 % of 725 samples. Because the elements varied together, we summarized the first principal component of the standardized instrument-reported values as a single elemental score per sample, which accounted for 39.2 % of their variation. Higher scores indicated more Ca, Mg, Na, S, Cl, Fe and Ti and less Si, indicating a contrast between Ca-, Mg-, Na-, S- and Cl-rich samples and Si-rich samples. Scores increased from west to east along the transect ( $\rho = 0.73$ ). Within sites, the elemental axis explained a small but significant share of compositional variation (partial RDA, 0.36 %,  $p = 0.010$ ; 0.31–0.54 % across sensitivity analyses), with similar effect sizes but weaker support when composition was measured as Bray–Curtis dissimilarity ( $p = 0.044$ – $0.056$ ).

118 genera correlated with the score, and they are mostly the same genera that changed with pH and transect position: alkaliphilic and halophilic Bacillota (*Polygonibacillus*, *Sediminibacillus*, *Aquibacillus*, *Halalkalibacter*, *Gracilibacillus*) were more abundant where scores were high, and *Steroidobacter*, *Caldilinea*, *Streptomyces*, *Roseisolibacter* and *Gemmatimonas* where scores were low (Supplementary Section S4.6.2). This overlap is expected, since the score itself rises steadily from west to east.

### **Exploratory models suggest an early richness response to rainfall**

Rainfall is rare in the Empty Quarter, and desert soil communities can respond within days of precipitation [30, 17]. Since the campaigns ran on fixed dates, samples were collected at different times after rainfall, which allowed us to explore whether recent rain was associated with alpha diversity. For each sample, we weighted rainfall over the preceding 60 days with a curve that peaks a set number of days before collection, and scanned peaks from 0.5 to 30 days for each of the three alpha diversity measures, accounting for

**Table 3.** Associations of environmental variables with transect position and community composition.

| Variable             | Transect $\rho$ | Genera | Within-site composition |
|----------------------|-----------------|--------|-------------------------|
| Temperature          | 0.98            | 112    | –                       |
| Relative humidity    | 0.92            | 111    | –                       |
| Rainfall             | 0.55            | 112    | –                       |
| Soil pH              | 0.82            | 117    | 0.28 % ( $p = 0.125$ )  |
| Elemental axis (XRF) | 0.73            | 118    | 0.36 % ( $p = 0.010$ )  |

site, compartment, campaign, and transect position. We repeated the analysis with two independent daily rainfall datasets: NASA POWER [31] and Open-Meteo [32] (Section ).

Both datasets selected rarefied richness at an early peak (Table 4, Figure S2). With NASA POWER, each additional millimetre of weighted rainfall was associated with 179 additional taxa, with the strongest weight 2 days before collection. With Open-Meteo, the estimate was 327 taxa per millimetre at 1 day. Rainfall explained a small share of the remaining variation in richness (3.6 and 3.8 %). When we refitted the full model on resampled sites, richness was selected in 72 and 68 % of refits and the peak remained within the first week, although the two datasets differed in effect size and exact timing.

Selecting the best of the 180 models inflates the strength of the association, so we calibrated the complete search against rainfall from other years, preserving the collection month and day, and the spatial pattern of the rainfall. With three complete years of daily data, only six such permutations were possible (smallest attainable value =  $1/6$ ). The observed association ranked first for NASA POWER ( $p = 1/6$ ), and second for Open-Meteo and both datasets combined ( $p = 2/6$ ).

Three additional checks described the timing and sensitivity of this exploratory association. First, we summed rainfall over fixed windows, which placed the largest positive estimate at 3–4 days with NASA POWER and at 1 day with Open-Meteo. Second, rainfall after collection produced a weak association with NASA POWER but a comparable richness association with Open-Meteo. Third, we removed individual ‘wet spells’, which identified the rainfall of 10 – 19 March 2023 as the main contributor to the association in both datasets (Supplementary Section S4.3).

**Table 4.** Association between antecedent rainfall and rarefied richness for two daily rainfall datasets.

| Dataset    | Peak (days)    | Effect (taxa mm <sup>-1</sup> ) | Variance explained (%) | Selected (%) | Calendar $p$ |
|------------|----------------|---------------------------------|------------------------|--------------|--------------|
| NASA POWER | 2.0 [1.5, 5.0] | 179 [64, 253]                   | 3.6 [0.2, 7.9]         | 72           | 1/6          |
| Open-Meteo | 1.0 [0.5, 6.0] | 327 [149, 797]                  | 3.8 [0.5, 8.7]         | 68           | 2/6          |

### Predicted functional potential follows geography and compartment

We predicted the functional potential of each sample from its 16S profile with PICRUSt2 [33] (Section ). Because prediction accuracy declines as sampled taxa become more distant from sequenced genomes, which is a particular concern for desert soils [34], we first assessed prediction quality. The median weighted nearest-sequenced-taxon index (NSTI) was 0.165, and was lowest in root-adjacent soil (0.134) and highest in shallow-subsurface soil (0.201; Supplementary Section S5.1). We then compared the predictions with KO profiles derived from metagenome-assembled genomes (MAGs) recovered from the collection of 150 shotgun libraries, using the 125 libraries with matched amplicon profiles. Across 3,471 shared KOs, predicted and MAG-derived profiles were strongly correlated within samples (median Spearman  $\rho = 0.74$ , 95 % CI 0.73–0.75) and between community-average profiles ( $\rho = 0.81$ ; Figure 4b). MAGs recruited on average 28.5 % of shotgun reads, so this comparison evaluates the predictions against the recovered fraction of each community (Supplementary Section S5.2).

Predicted metabolism was dominated by core cellular functions. Of the 462 predicted MetaCyc pathways, biosynthesis pathways accounted for 66.8 % of predicted pathway abundance, energy and central metabolism for 17.2 %, and degradation and utilization for 10.7 %. Aerobic respiration (cytochrome *c*) was the most abundant pathway (1.7 %), followed by branched-chain amino acid and fatty acid biosynthesis. Carbon fixation pathways were of low abundance, where the Calvin–Benson–Bassham cycle ranked 44th (0.6 %) and the reductive TCA cycle 122nd (0.4 %), consistent with the low relative abundance of Cyanobacteriota (0.4 %; Supplementary Section S5.3).

Predicted pathway profiles showed the same geographic and compartment patterns as community composition. Transect position explained 23.3 % of the variation in pathway composition among sites (redundancy analysis; 95 % CI 15.4–31.3 %; 21.3–36.1 % across sensitivity analyses), and 92 of the 200 pathways were correlated with transect position ( $q < 0.05$ ). Among them, pathways for cofactor and nucleotide biosynthesis, such as queuosine and pantothenate biosynthesis, increased eastward ( $\rho = 0.73$ – $0.75$ ), whereas pathways for sugar fermentation and degradation, such as the *Bifidobacterium* shunt and galactose degradation, decreased ( $\rho = -0.59$  to  $-0.72$ ; Supplementary Table S27).

Pathway composition also differed among compartments (pseudo- $F = 6.37$ ,  $p = 0.0001$ ; 172 site visits). As for community composition, the most consistent differences were between root-adjacent soil and both bulk compartments (standardized displacement 0.45 and 0.43;  $q = 0.0002$ ; Figure 4a), whereas the difference between shallow-subsurface and surface soil (0.33,  $q = 0.019$ ) was not significant when Trip 3 was omitted. Of 600 pathway-compartment contrasts, 270 were significant ( $q < 0.05$ ). Compared with both bulk compartments, root-adjacent soil was enriched in pathways for glucose and xylose, fructuronate and 4-aminobutanoate degradation, polyamine and thiamin biosynthesis, and formaldehyde assimilation, but depleted in starch degradation (Supplementary Table S24).

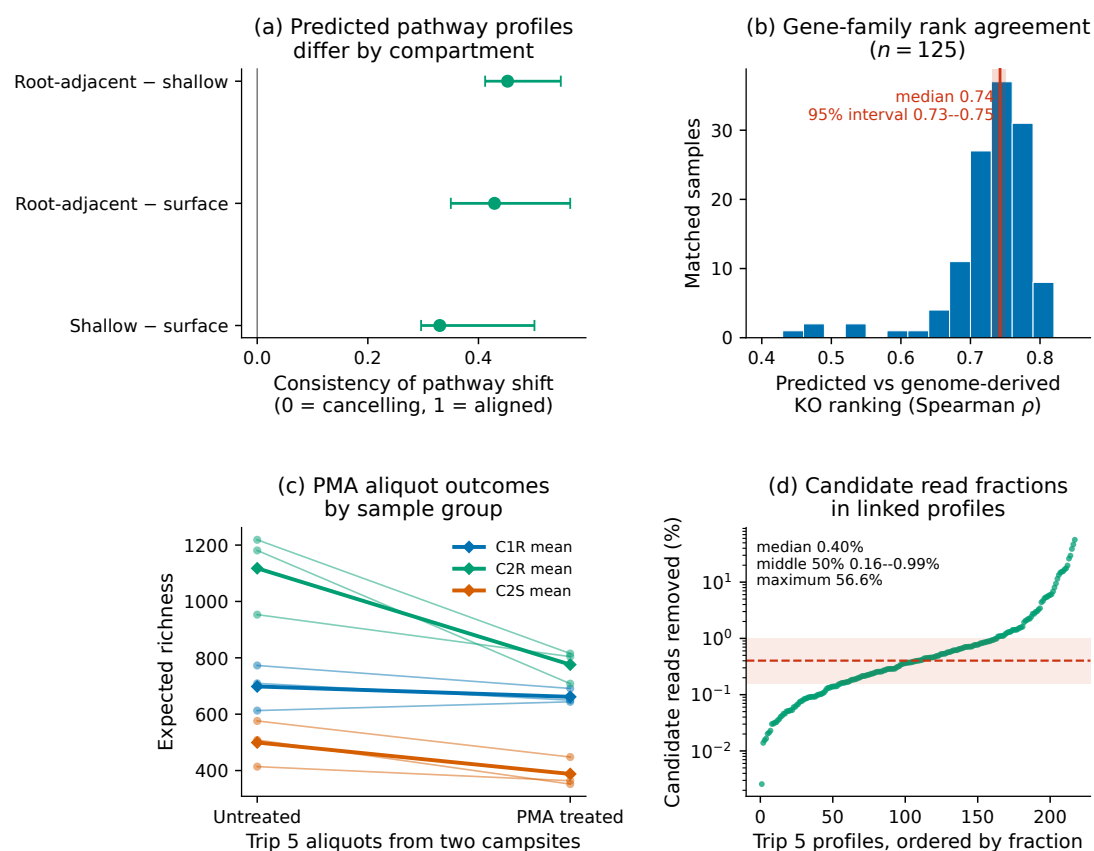
We examined mechanisms relevant to survival in hyperarid soil by screening 20 marker gene sets in the MAGs and the PICRUSt2 predictions (Table 5). Marker genes related to CO oxidation and hydrogen metabolism were abundant, as 38.5 % of MAGs carried the aerobic CO dehydrogenase gene *coxL* and 22.6 % a [NiFe]-hydrogenase (membrane-bound group 1 or soluble group 3d), mostly within Actinomyce-tota, whereas the nitrogenase gene *nifH* was only in 1.0 %. Carrier abundance varied descriptively among compartments: hydrogenase carriers were most abundant in shallow-subsurface soil, carotenoid and pho-tolyase carriers in surface soil, and sporulation carriers in root-adjacent soil. Along the transect, predicted trehalose and carotenoid synthesis genes decreased eastward, whereas ectoine synthesis, sporulation, and superoxide dismutase genes increased. Predicted and MAG-derived abundances agreed for most stress response, osmolyte, and trace-gas markers ( $\rho = 0.51$ – $0.76$  across 119 samples), but poorly for genes that are nearly universal (*uvrA*, *recA*) or rare (*nifH*, *psbA*, *nosZ*; Supplementary Section S5.4).

**Table 5.** Selected marker-gene sets in MAGs and PICRUSt2 predictions. MAGs: percentage of 975 annotated MAGs carrying at least one KO in the set. Su, Sh, Ro: mean relative abundance (%) of carrier MAGs in surface, shallow-subsurface and root-adjacent shotgun libraries. Transect  $\rho$ : Spearman correlation of predicted abundance with transect position. Agreement  $\rho$ : Spearman correlation between predicted and MAG-derived abundance across 119 libraries. \*  $q < 0.05$ . All 20 sets are given in Supplementary Tables S25 and S26.

| Marker set                              | MAGs (%) | Su   | Sh   | Ro   | Transect $\rho$ | Agreement $\rho$ |
|-----------------------------------------|----------|------|------|------|-----------------|------------------|
| CO dehydrogenase ( <i>coxL</i> )        | 38.5     | 43.8 | 45.5 | 40.4 | −0.11           | 0.71             |
| [NiFe]-hydrogenase                      | 22.6     | 17.2 | 31.9 | 15.0 | 0.05            | 0.74             |
| RuBisCO ( <i>rbcL/cbbL</i> )            | 24.6     | 18.2 | 24.1 | 18.1 | 0.26            | 0.33             |
| Nitrogenase ( <i>nifH</i> )             | 1.0      | 0.5  | 0.3  | 0.8  | 0.08            | 0.04             |
| Trehalose ( <i>otsA</i> , <i>otsB</i> ) | 72.7     | 85.6 | 89.0 | 83.8 | −0.71*          | 0.53             |
| Ectoine ( <i>ectA–ectC</i> )            | 36.4     | 25.9 | 28.0 | 31.3 | 0.36*           | 0.65             |
| Carotenoids ( <i>crtB</i> )             | 49.3     | 60.6 | 52.7 | 53.4 | −0.59*          | 0.13             |
| Sporulation ( <i>spo0A</i> )            | 7.3      | 5.1  | 4.3  | 7.5  | 0.56*           | 0.51             |
| Superoxide dismutase ( <i>sodA</i> )    | 82.8     | 68.4 | 67.4 | 78.8 | 0.70*           | 0.66             |
| Dps ( <i>dps</i> )                      | 65.1     | 58.2 | 47.3 | 69.5 | 0.43*           | 0.76             |

### PMA-accessible DNA contributed to measured richness

To assess the contribution of PMA-accessible DNA, we treated paired aliquots of three parent soil samples from two Trip 5 campsites with propidium monoazide (PMA), which suppresses amplification of extracellular DNA and DNA in membrane-compromised cells (Supplementary Section S3). PMA-treated aliquots had lower rarefied richness than untreated aliquots in eight of the nine pairs (Figure 4c). The mean reduction was 37 ASVs in root-adjacent soil from the first campsite, 342 in root-adjacent soil and 111 in surface soil from the second campsite. Shannon diversity varied less consistently (+0.73, −0.02 and −0.31). Each group comprised three slurries prepared from one parent soil tube, so these comparisons describe the PMA response of those three parent samples.



**Figure 4.** Predicted function, genome-derived agreement and assay controls. (a) Consistency of predicted-pathway compartment shifts, with 95 % intervals from resampling whole sites. (b) Agreement between predicted and recovered-genome-derived gene-family rankings in 125 matched profiles; red marks the median and its 95 % site-bootstrap interval. (c) Expected richness before and after PMA treatment in nine aliquot pairs, grouped as C1R, C2R and C2S at two campsites. Each group comprises three separately prepared slurries from one parent soil tube; lines join the treated and untreated aliquots of each slurry. (d) Fraction of reads assigned to candidate contaminants in 217 batch-linked Trip 5 profiles; red marks the median and interquartile range.

## DISCUSSION

### The Empty Quarter microbial ecology

The microbial communities of the Saudi Empty Quarter are organized both across hundreds of kilometres and within a few centimetres of soil. Taxon composition varied along the sampled transect, mostly through replacement of genera between sites, and compartments differed consistently in composition and alpha diversity within sites. Long term climate, soil pH, and soil chemical composition covaried with the geographic pattern, and predicted metabolic potential followed the same geographic and compartment structure. Together, these observations describe a recurring microbial mosaic at landscape and local scales.

Phyla abundances in the Empty Quarter share similarities and differences with other arid deserts. Empty Quarter soils are enriched in Pseudomonadota, Bacillota, Actinomycetota and Chloroflexota, whereas Acidobacteriota and Verrucomicrobiota, which are prominent in temperate and agricultural soils, were rare [3, 27, 5]. At the genus level, the dominance of *Domibacillus*, *Bacillus* and other spore-forming Bacillota, together with the low relative abundance of Cyanobacteriota, distinguishes these open sand soils from biocrust-dominated desert surfaces.

Across the landscape, community dissimilarity increased with geographic distance in all sampled compartments, and turnover accounted for most of the dissimilarity between sites. Distant sites therefore hosted largely distinct assemblages rather than progressively poorer subsets of a single community. Distance decay has also been reported in Negev soils [35], and Namib dune systems show habitat-specific community structure across dune zones [13]. In the Empty Quarter, the change along the transect was not a simple west to east gradient; a quadratic term substantially improved the fit. Nestedness increased with distance in bulk soils, consistent with the lower alpha diversity of the eastern sites. Because of residual spatial autocorrelation, the strength of this transect trend should be read as a description of the sampled sites rather than a test of the gradient. The pattern was not driven by the saline pans and oases interspersed along the transect, as it persisted among dune sites alone.

The geographic pattern coincided with gradients in climate, pH, and chemical composition. The eastern end of the transect was the warmest, most humid, and most alkaline, and was enriched in alkaliphilic and halophilic Bacillota such as *Halalkalibacter*, *Aquibacillus* and *Sediminibacillus*, whereas Actinomycetota such as *Cellulomonas* and *Marmoricola* declined eastward. Soil pH shared a substantial association with transect position in site-average composition, while its unique and within-site associations were small. This is consistent with the role of pH as one of the main correlates of soil bacterial composition across biomes [36]. The elemental axis, contrasting Ca-, Mg-, Na-, S- and Cl-rich samples with Si-rich samples, explained less than 1 % of compositional variation within sites. Alpha diversity decreased with long term temperature, humidity, and rainfall, which runs counter to the expectation that more moist desert soil supports more diverse communities [37]. Because these climate variables, pH and elemental composition all change along the same transect, our design cannot separate their individual effects, and the associations identify a shared environmental gradient rather than its drivers.

Precipitation may also affect community composition in short time scales. Exploratory models selected a positive association between rarefied richness and rainfall in the first days before collection using two datasets. Pulse-reserve theory predicts that rare rainfall can briefly release water limitation before drying restores it [16, 17]. The comparable post-collection association in Open-Meteo and dependence on a single wet spell in March 2023 make independently sampled storm events necessary to test the timing of this response. Sampling before rain and during the first days of drying, together with measurements of cell abundance and activity, would distinguish growth from the detection of previously dormant cells.

Within sites, compartments varied consistently in composition, with the largest and most consistent differences between root-adjacent soil and both bulk soil compartments. Vertical stratification has also been reported in an Atacama soil profile and between surface and subsurface Mojave biocrust communities [14, 10]. Surface soil was enriched in pigmented, desiccation-tolerant genera such as *Deinococcus* and *Rubrobacter*, and shallow-subsurface soil in nitrifiers such as *Nitrospira* and MND1. Root-adjacent soil was enriched in plant-associated genera, including *Neorhizobium*, *Devosia* and *Pseudomonas*, whereas the root-nodule genus *Bradyrhizobium* was depleted. Root-adjacent soil had the lowest alpha diversity per sample, higher dispersion among sites than surface soil, the largest core of shared genera, and slower distance decay than shallow-subsurface soil. Individual root-adjacent samples therefore contained smaller and less even subsets of a regional pool that was more widely shared across the landscape than the pools of bulk soil. Studies that identified desert plants have found host- and compartment-specific recruitment

of root-associated bacteria [38, 6, 15, 39]. Our root-adjacent compartment includes soil near and attached to the roots of unidentified plants, so these patterns describe the compartment as sampled. Sampling identified hosts at measured distances from roots would connect them to host identity and rhizosphere processes.

Predicted pathways followed the same geographic and compartment structure as taxonomic composition. Pathways for cofactor and tRNA-related biosynthesis increased towards the east, whereas pathways for sugar fermentation and degradation decreased. Near roots, pathways for glucose and xylose degradation, polyamine and thiamin biosynthesis, and formaldehyde assimilation were enriched, whereas starch degradation was depleted. The enrichment of sugar degradation and formaldehyde assimilation is consistent with the use of root-derived sugars [40], and of plant-derived methanol [41]. The genome screen adds candidate mechanisms: over a third of the analysed MAGs carried CO dehydrogenase markers and about a fifth carried [NiFe]-hydrogenase markers, and hydrogenase carriers were most abundant in shallow-subsurface soil. Trace-gas oxidation can sustain bacteria in hyperarid soils [42], and direct measurements of gas uptake, ATP and respiration in the Namib show how such capacities can be evaluated after wetting [43]. Because the hydrogenase markers include several enzyme groups and the predictions rely on reference genomes, sequence-resolved classification and process measurements are needed to confirm these mechanisms.

PMA treatment reduced rarefied richness in most paired samples, indicating that PMA-accessible DNA contributed to measured richness in the sampled soils. Because aliquots came from single soil samples, these comparisons do not extend to the rest of the survey, and PMA efficiency depends on the soil matrix, biomass, and community composition [44]. Candidate contaminants identified from control samples accounted for a small fraction of reads, and their removal preserved alpha-diversity rankings.

## **A regional framework for the Arabian Peninsula**

Earlier studies in the Arabian Peninsula established habitat-specific patterns at individual locations. An inland sabkha in the United Arab Emirates hosted a saline microbial mat [25], and studies in Saudi Arabia examined highland soils [21], arid sites [22], soil crusts [45], and the root microbiomes of desert plants [6]. Our survey contributes an additional scale to this work; a landscape-scale transect with repeated sampling of three soil compartments over five campaigns.

We found that soil in the open Empty Quarter is alkaline, which falls within the range reported for soils in Saudi Arabia [22]. At AlUla, soil beneath fertility islands formed by desert plants hosted bacterial communities distinct from those in the surrounding bulk soil, which was also more alkaline [46]. In the Empty Quarter, pH increased from west to east, and its association with community composition was largely shared with geography. The small unique pH component and strong geographic covariance support a landscape-scale association with the sampled environmental gradient. Paired sampling of patches with vegetation and surrounding bulk soil would test whether local pH differences, such as those observed in AlUla, also shape composition at the scale of individual plants.

Regional experiments also provide practical models for follow up studies. In soil crusts associated with *Rhazya stricta*, biomass increased threefold within a week of watering [45], consistent with the early response window selected by our rainfall models. In Jizan and AlWahbah, host-resolved surveys found the strongest plant-associated differentiation in the root endosphere [6]. Our root-adjacent compartment, which differed from bulk soil and had the lowest alpha diversity, represents the soil end of this root–soil gradient. Post-rainfall sampling and comparisons among identified host plants would connect these regional observations.

Our survey also provides a baseline for biodiversity monitoring in restoration in the region, and three of our results inform monitoring design. First, because turnover accounted for most of the dissimilarity between sites, no single site represents the regional community, and monitoring should retain broad geographic coverage. Second, because the three compartments differed in composition and alpha diversity, they should be sampled separately. Third, because root-adjacent communities had higher dispersion among sites than surface soil, they may require more replicates than bulk soil to detect change. Around rainfall events, the first days after rain offer a candidate window for additional sampling. The linked samples, environmental measurements, sequences, and analysis inputs make these comparisons reusable [26].

### **Sampling constraints and tests of mechanism**

All sites lie on a single west–east transect, along which climate, pH and soil chemical composition change together, and every campaign proceeded in the same direction. Geographic position is therefore confounded with collection order and time of day, and the environmental gradients cannot be separated from one another or from geographic distance. In addition, community composition remained spatially autocorrelated after accounting for the transect, and support for the transect trend depended on the assumed spatial covariance and independent-noise fraction. We therefore treat the transect model as a description of the sampled landscape. Future surveys could reverse the direction of travel between campaigns, add shorter transects crossing the main route where terrain allows, and install autonomous soil sensors at selected sites. These changes would help separate geographic, environmental, and collection time effects while retaining feasible routes through the dune interior [7].

The environmental measurements describe the soil samples indirectly. Climate summaries describe air, rather than soil, conditions, pH was measured on stored soil samples with uneven coverage of campaigns, and XRF measures total elemental composition rather than soluble ions or salinity [47]. Soil moisture, temperature, conductivity, and soluble-ion measurements would characterize water and salt exposure more directly. Similarly, the exact sampling depth and distance from root were not recorded, and host plants were not identified, so the compartments are defined by how samples were collected. Recording these fields would help separate plant-associated differences from shared habitat effects.

Functional analyses describe metabolic potential rather than activity. PICRUST2 predictions depend on how closely the sampled taxa resemble sequenced genomes, and the MAGs used to evaluate them mapped to a small fraction of whole-genome reads. Sequence-resolved validation of marker genes and functional profiles derived directly from metagenomic reads would extend this comparison to a larger fraction of each community. Measurements of proteins, metabolites, cell abundance, and process rates, such as trace-gas uptake, would connect the predicted mechanisms to microbial activity [48, 43].

Control samples did not cover all campaigns and sample processing stages. Extraction blanks were only linked to specific batches, and the recovery of mock community controls was variable. Future surveys should sequence extraction and PCR blanks with every batch, include positive standards, and record DNA concentrations and pooling measurements alongside sequencing metadata [49].

### **Conclusions**

The Empty Quarter hosts a structured microbial landscape. Bacterial assemblages change across the west to east, mainly through turnover of genera, and compartments differ in composition and alpha diversity within sites. Climate, soil pH, and soil chemical composition covary with the geographic pattern, and predicted metabolic potential also follows the same geographic and compartment structure. Exploratory rainfall models identify the first days after rain as a window for event-based sampling. Together, the survey provides a regional baseline and specific hypotheses on geographic, plant-associated, and metabolic patterns for repeated surveys and experiments.

## **MATERIALS AND METHODS**

### **Study design and sample processing**

#### ***Field design and sample collection***

We sampled a transect through the Empty Quarter during expeditions on March 17–20, 2023 (T1), July 14, 2023 (T2), February 17–21, 2024 (T3), August 29–31, 2024 (T4), and October 12–15, 2025 (T5). The repeated transect comprised 60 sites, except T2, which was restricted to 8 sites due to extreme heat. At each site, we sampled bulk surface soil (0cm), bulk subsurface soil (nominal depth 5–10cm) and subsurface soil referred to as “root-adjacent” collected near plant roots (5–10cm), collecting three replicates per compartment. The exact depth of each bulk subsurface sample was not recorded. The sampling initially targeted the rhizosphere, but most plants were so desiccated that the soil attached to the roots could not be obtained; therefore, we collected the soil as close to the root as possible, which in some cases included the soil attached to the roots. Soil was placed in sterile tubes by scooping directly with the tube or with a sterile plastic spoon. Samples were kept at  $-20^{\circ}\text{C}$  in a portable freezer during each campaign and transferred to  $-80^{\circ}\text{C}$  in the laboratory.

Four sites sampled only during T1 contributed 10 additional samples. We retained them for description but excluded them from repeated-site analyses because later expeditions could not revisit these sites. The analysis-specific cohorts are defined in the corresponding Methods and supplementary sections.

### **Sample processing**

We extracted 0.9–1.2 g of soil with QIAGEN PowerLyzer PowerSoil or DNeasy PowerSoil Pro kits, homogenizing with a PowerLyzer 24 (4,000 rpm, one 45-second cycle), processing manually or with a QIAcube Connect, and eluting in 50  $\mu$ L. We amplified the V3–V4 region of the 16S rRNA gene with the Bakt\_341F (CCTACGGGNGGCWGCAG) and Bakt\_785R (GACTACHVGGGTATCTAATCC) primers [50], both carrying Illumina overhang adapters. Amplification followed a two-step protocol [51] where we performed a first PCR with TaKaRa Ex Taq DNA Polymerase (cat. no. RR001B), and checked products on a 1 % agarose gel for amplicons of the expected size (500 bp including overhangs). We re-amplified the samples without visible products with more input DNA (8  $\mu$ L, then 12  $\mu$ L), and samples without a product after three attempts were classified as failed.

We indexed products with the QIAGEN Multiplex PCR Kit, and purified indexed libraries with Agencourt AMPure XP beads, and quantified them with the Qubit high-sensitivity double-stranded DNA assay. We checked the final library sizes on an Agilent TapeStation 4200 and pooled them according to their molarity. Pools were sequenced on an Illumina NovaSeq 6000 SP flow cell with a 500-cycle kit to generate  $2 \times 250$ -bp paired-end reads. Additional details on protocols and sample preparation are given in Supplementary Section S1.2. Sample processing and library preparation were performed in the PI's laboratory and sequencing was conducted at the Bioscience Core Lab at KAUST.

### **Read processing**

We processed reads with nf-core/Ampliseq v2.14.0 [52, 53], and used DADA2 to infer exact amplicon sequence variants (ASVs) and SILVA v138.2 [54] to assign taxonomy. The resulting count table contained 1,271 profiles, including 24 control profiles (18 extraction blanks and six PCR no-template controls), and 351,472 ASVs. We assessed contamination at different study stages following recommendations for low-biomass microbiome studies [49] (Supplementary Section S2, Table S2). We excluded control samples and 10 samples that had < 1,000 reads, leaving 1,237 samples.

### **Relic-DNA removal experiment**

PMA treatment suppresses amplification of accessible DNA, including extracellular DNA and DNA in membrane-compromised cells [44, 55, 56]. We prepared three separate soil slurries from each of three parent soil tubes collected at two T5 campsites: root-adjacent soil at C1 and C2, and surface soil at C2. Each slurry was homogenized in sterile phosphate-buffered saline (PBS, pH 7.4) at a 1 : 10 (w/v) ratio and split into treated and untreated aliquots. We added PMAxx (Biotium, Hayward, USA) to treated aliquots at a final concentration of 50  $\mu$ M. Both aliquots underwent a 10 min dark incubation, photoactivation with a 450 nm LED lamp for 35 min on a white reflective surface (15–20 cm from the lamp), and rotation at 15 min intervals. We stored samples at  $-20^{\circ}\text{C}$  until DNA extraction, which was performed using 860  $\mu$ L from each aliquot with the DNeasy PowerSoil Pro kit (QIAGEN). We amplified the DNA and sequenced the samples with the protocols described in Section .

We calculated Hurlbert expected richness at 123,897 reads, the minimum depth among the 18 libraries, and Shannon diversity at full read depth. We report each treated-minus-untreated difference and the mean difference within each of the three groups. Supplementary Section S3 gives the protocol and endpoint calculations.

### **Environmental measurements**

#### ***Weather and climate***

During our first campaign, we recorded the geographic coordinates of each sampling site along the transect and returned to these exact coordinates in all subsequent campaigns. At each site visit, we recorded air temperature and, where available, atmospheric pressure and relative humidity. We also photographed the sampling site and the plant from which we collected root-associated soil. We documented environmental features at each site and used the features to assign one of the following landforms: sand dune, saline pan, oasis, gravel plain, dune slack, aeolian lake, and oil spill.

Since the environmental measurements we obtained capture weather at a point in time, we supplemented them with climate data from external sources. For each site, we retrieved monthly air temperature, total rainfall, and relative humidity from the Open-Meteo historical API resource [32]. We also obtained daily precipitation for each site from the NASA POWER PRECTOTCORR resource [31], using complete calendar days preceding the sampling date.

### **Soil measurements**

We measured pH from 767 stored soil samples from all campaigns. We suspended air dried soil sieved to  $\leq 2$  mm in 0.01 M  $\text{CaCl}_2$  at a 1:2.5 (w/v) ratio, mixed on a hula mixer for 30 min, and left them to stand for 30 min. We measured pH using an Accumet AB315 meter fitted with a Fisher Scientific Accumet electrode (cat. no. 13-620-112), which we calibrated at the start of each session and rinsed with Milli-Q water between measurements. We retained only measurements with a valid calibration defined as an electrode slope of 95–102 %, and a pH 10 buffer read-back of 9.9–10.1 and a value within the physical range. pH coverage was incomplete and non-random across campaigns (Supplementary Section S1.4).

In the laboratory we measured total elements in 725 stored soil samples from all campaigns using a Vanta Handheld XRF Analyzer in Fast Mode. A separate in-situ field XRF workflow (only in T5) was used as a descriptive diagnostic and is reported in Supplementary Section S1.5. We retained the 11 elements detected in at least 75 % of samples and applied a  $\log(1+x)$  transformation. We standardized the values within each campaign to reduce between-campaign differences. We summarized the standardized element values by principal component analysis (PCA) and used the first component as a single elemental score per sample.

### **Microbial community analysis**

#### **Community composition**

To describe the overall composition of the microbial communities, we summed ASV counts at the phylum, class, and genus levels (SILVA v138.2, Section ) and calculated the mean relative abundance of each taxon across the 1,227 samples. We retained unclassified reads. We summarized composition across all samples, by compartment, and by transect third (20 sites each, ordered along the transect coordinate). Full tables are given in Supplementary Section S4.1.

#### **Alpha diversity (within-sample)**

We summarized within-sample diversity using three measures. We calculated Shannon diversity ( $H$ ) with natural logarithms at full read depths as it is less sensitive to variation [57]. We also estimated rarefied richness as Hurlbert's expected number of ASVs as it controls for variation in sequencing depth [58] at 25,000 reads ( $E[S_{25k}]$ ) near the 5th percentile of library size (median 382,432 reads per sample). This allowed us to retain 95 % of the samples, and the rest were omitted from richness comparisons. To measure abundance relative to rarefied richness, we used a normalized Shannon index  $H/\log(E[S_{25k}])$  (Supplementary Section S4.2).

**Compartment contrasts.** To separate the effect of compartment from differences among sites and campaigns, we compared surface, shallow subsurface, and root-adjacent soil with paired, site-level differences in a repeated measures design that removes within-site and within-campaign variation. For each compartment contrast, we used only the sites and campaigns for which we had samples, averaged the differences within campaign, then averaged across campaigns within each site. We obtained confidence intervals by site-level bootstrap and tested the paired differences with the Wilcoxon signed-rank test, controlling the false discovery rate separately for the Shannon and normalized Shannon comparisons. To check whether sequencing depth and DNA extraction kit affected the differences, we refitted the contrasts with Gaussian generalized estimating equations (GEEs) that accounted for repeated sampling of the same site (Supplementary Section S4.2.2, S4.2.3).

**Climate associations.** We tested whether alpha diversity varied with weather and climate measurements. Associations with our field weather measurements did not remain after adjustment for campaign and transect position and correction for multiple testing, so we conducted the remaining analyses with external long-term climate datasets (Section ). We averaged each diversity index within site and used Spearman's correlation with 49-month means of temperature, rainfall, and humidity. We corrected the resulting tests as one family with Benjamini-Hochberg correction, and derived intervals by paired-site bootstrap resampling. Because the three climate summaries are themselves ordered along the sampling transect, we interpret these correlations as descriptions of the environmental axis along which diversity changes, rather than a means of identifying a causal driver (Supplementary Section S4.2.4).

**Short-term rainfall.** To test whether samples collected soon after rainfall were more diverse, we modeled alpha diversity measures as a function of time-dependent cumulative rainfall over the 60 days preceding collection under a rise-and-decay distributed lag kernel (Supplementary Section S4.3.1). Because the lag was unknown, we fitted this model across lags from 0.5 to 30 days for each diversity index, adjusting for site, compartment, campaign, and transect position. We repeated this analysis with

two independent rainfall datasets (Section ). Since selecting the lag with the strongest effect inflates significance, we assessed the whole search against a randomization null that preserves the amount and spatial pattern of rainfall. We estimated the selected effect by refitting the full model on site-level bootstrap resamples. Detailed methods are described in Supplementary Section S4.3.

#### **Beta diversity (between-sample)**

For beta diversity analyses, we summed replicates for each site  $\times$  campaign  $\times$  compartment combination into a single genus count profile, and discarded profiles with  $< 2,000$  reads, yielding 630 profiles across 60 sites. We kept the 200 genera with highest mean relative abundance present in at least 20 % of profiles. We centre log ratio (CLR) transformed the profiles after adding a 0.5 pseudocount for compositional analyses. We calculated Aitchison distance, the Euclidean distance between CLR profiles, which compares taxa through their abundance ratios [59]. We treated site as the independent unit throughout as we applied permutations to whole sites, and obtained confidence intervals by delete-one-site jackknife or site-level bootstrap (Supplementary Section S4.4.1).

**Transect model (Redundancy analysis).** To test whether community composition changed across the study transect, we first removed campaign and compartment effects by subtracting from each profile the mean CLR profile of its campaign  $\times$  compartment combination (mean taken across sites), and averaged the centred profiles to a single profile per site. The transect coordinate was defined as the first principal component (PC1) of site coordinates in kilometers, oriented from west to east. We related composition of site profiles to linear and quadratic transect terms by redundancy analysis (RDA) [60] and summarized the fit as variance explained, with a delete-one-site jackknife interval, and tested the fit with 999 site permutations (Supplementary Section S4.4.2). Since the residuals remained spatially autocorrelated (multivariate Moran's  $I$ , 5-nearest neighbor graph), we treat the variance explained as descriptive and report its sensitivity to spatially correlated errors (Supplementary Section S4.4.3). We also compared site-level CLR of each genus with the transect coordinate (200 tests, Benjamini–Hochberg).

**Distance decay.** We estimated distance decay for each compartment with Aitchison distance. We used the same campaign  $\times$  compartment centring, but averaged profiles within site *and* compartment. Within each compartment, we regressed pairwise Aitchison distance on geographic distance using multiple regression on distance matrices (MRM). The resulting linear regression slope gives dissimilarity per 100 km. Because pairwise distances involving the same site are not independent, we assessed significance by permuting site labels (9,999 permutations) rather than individual distances. We applied each permutation to all three compartment matrices at once, keeping geographic distance fixed, so that differences between compartments could be tested against the same null.

To determine whether spatial turnover rates varied among compartments, we compared decay rates using differences between distance matrices (*surface – root-adjacent* and *shallow subsurface – root-adjacent*). We first tested slope differences with an omnibus test, then corrected the two contrasts with maximum-statistic correction. Finally, to account for the spatial structure and non-independence of the matrix elements, confidence intervals for the regression slopes were calculated via a delete-one-site jackknife resampling, omitting each site in turn (Supplementary Section S4.4.4).

**Compartment differences.** We compared compartments in campaign  $\times$  site blocks. Within each block, we subtracted the block's mean CLR profile from each compartment profile to remove site, campaign, and site  $\times$  campaign effects, and averaged centred blocks within site. We tested for overall differences among compartments with a PERMANOVA on blocks that had samples in all compartments, resulting in 170 blocks across 60 sites. We computed a pseudo- $F$  statistic [61] on block-centred CLR profiles with permutations restricted within sites (999). As this is sensitive to differences in both location and dispersion [29], we tested pairwise differences in location using paired sign flip permutation tests (999) on the mean within-site difference vector. We used every block containing both compartments of that pair. We summarized the consistency of each pairwise shift across sites as a standardized displacement (Supplementary Section S4.4.5).

To test whether compartments differed in their community variability, we compared multivariate dispersion. We measured the Aitchison distance of each profile to the centroid of its campaign  $\times$  compartment combination and averaged these distances within site, so that each site contributed one dispersion value per compartment regardless of how many times it was sampled (Supplementary Section S4.4.5). We then tested mean paired difference for each pair of compartments with a sign flip permutation test (999). For each of the 200 genera, we tested the three pairwise compartment contrasts with the same paired sign-flip test. We applied Benjamini–Hochberg correction separately to the pairwise location, dispersion,

and genus-level contrasts.

### ***Turnover and nestedness***

Compositional differences between sites can arise because different taxa replace one another (turnover) or because communities contain subsets of taxa found in richer ones (nestedness). This partition depends on taxon presence or absence, which CLR profiles cannot represent. We used presence/absence data from all genera, and pooled raw counts within each site and compartment across campaigns, obtaining one profile per site  $\times$  compartment. We rarefied each pooled profile to the smallest pooled library (12,865 reads). We partitioned Sørensen dissimilarity into turnover (Simpson dissimilarity) and nestedness components [28]. We regressed each component on geographic distance, and tested significance with joint site permutations. To compare the size of the shared taxon pool among compartments, we used the same pooled counts, rarefied 20 times to 12,865 reads, and defined the core community as the genera present at  $\geq 90\%$  of sites in each compartment (Supplementary Section S4.5).

### ***Composition and the environment***

**Soil pH and XRF elements.** To test whether soil pH and elemental composition were associated with microbial community composition, we used partial RDA on the CLR profiles of all site  $\times$  campaign  $\times$  compartment combinations with a corresponding measurement (560 profiles for pH, 621 for XRF elements). We entered each variable after campaign, compartment, and site, to prevent the association from simply reproducing the west to east gradient. Because repeated profiles from one site are not independent, we restricted permutations within site (999 permutations). We partitioned pH and geographic associations using one fixed CLR response and a common variance denominator. We adjusted the response, pH and linear and quadratic transect terms for campaign-by-compartment categories, with equal total weight per site, and averaged the residuals within each site. Four models of these 60 site averages (intercept only, pH, geography, and both) separated unique pH, unique geography, shared and unexplained fractions. We also fitted the corresponding weighted models to all 560 groups and assessed sensitivity to site omission and spatial-block resampling (Supplementary Section S4.6.1).

For the XRF elements, we used the same profiles and repeated the test on Bray–Curtis dissimilarities (distance-based RDA) to check that the result did not depend on the CLR transformation (Supplementary Section S4.6.2). At the genus level, we correlated each genus's site-level CLR profile with mean pH at the site and with the elemental axis (Spearman), and corrected with Benjamini–Hochberg.

**Landforms.** Because landforms such as saline pans and oases are unevenly distributed along the transect, we explored whether landform could explain community composition. We compared the 44 sand dune sites and 9 saline pan sites in each alpha diversity measure (Section ) with Mann–Whitney tests and summarized the 3 oases sites descriptively. We also compared their composition with PERMANOVA on site-level CLR profiles with 9,999 permutations (Section ) before and after removing the linear and quadratic effects of transect position, and tested individual genera with Mann–Whitney tests (200 tests, Benjamini–Hochberg). To test whether the west to east patterns persisted within a single landform, we refitted the transect RDA, the genus–transect correlations and the climate–diversity correlations on sand dune sites only (Supplementary Section S4.6.3).

### ***Predicted functional potential***

#### ***Function prediction***

We predicted the functional potential of each sample from the ASV table with PICRUSt2 v2.4.1 [33], which infers gene family content from the nearest sequenced reference genomes. We obtained unstratified abundances of KEGG orthologs (KOs) [62] and MetaCyc pathways [63] (462 pathways) for all 1,227 samples. Because prediction accuracy declines as sampled taxa become more distant from sequenced genomes, a particular concern in under-studied desert soils [34], we assessed prediction reliability with the weighted nearest-sequenced-taxon index (NSTI), where smaller values indicate less extrapolation from reference genomes.

#### ***Comparison with metagenomes***

We used a catalogue of 991 metagenome-assembled genomes (MAGs) recovered from 150 Trip 5 shotgun libraries in the companion study. Of these genomes, 990 had KO annotations from eggNOG-mapper v2.1.12 [64]. For each shotgun library, we constructed a community KO profile by weighting the genome KO annotations by genome relative abundances estimated with CoverM [65]. We compared the 125 shotgun libraries with matched amplicon profiles across 3,471 shared KOs, using Spearman rank

correlations within each matched library pair and between the community-average profiles, with intervals from a site-level bootstrap (Supplementary Section S5.2).

### ***Pathway composition***

We tested whether predicted functional potential retained the patterns observed in taxonomic composition. We analyzed pathway abundances with the same design as the genus-level analyses (Section ). We summed replicates within each site  $\times$  campaign  $\times$  compartment combination (633 grouped profiles) and retained the 200 pathways with the highest mean relative abundance present in at least 20% of profiles. We CLR transformed their abundances after adding a pseudocount of 0.5. We tested spatial distribution with the transect RDA and compartment differences with PERMANOVA and paired sign-flip tests, as described for taxonomic composition (Section ), using 9,999 permutations throughout. Compartment tests used the 172 campaign  $\times$  site blocks in which all three compartments were sampled. To identify the pathways driving these patterns, we tested each pathway across the three compartment contrasts with paired sign-flip tests, and correlated each pathway's site-level CLR value with the transect coordinate (Spearman), applying Benjamini–Hochberg correction within each family.

To describe the dominant predicted metabolism, we calculated the mean relative abundance of each pathway across the 1,227 samples and grouped pathways into biosynthesis, degradation or utilization, energy and central metabolism, and other, based on keywords in their MetaCyc descriptions (Supplementary Section S5.3).

### ***Marker genes for desert survival***

We examined traits relevant to survival in hyperarid soil by defining 20 marker gene sets covering trace gas oxidation, carbon fixation, nitrogen cycling, osmolyte synthesis, pigments, sporulation, DNA repair and oxidative stress defence, each consisting of one or more KOs. We considered a genome a carrier when it contained at least one KO in the set. For each set, we calculated the fraction of the 975 annotated MAGs that carried it and, in each shotgun library, the relative abundance of those carrier genomes in the community. In parallel, we calculated the predicted abundance of the same KOs as a share of total predicted KO abundance in the PICRUSt2 profiles. We compared compartments using paired site-level  $\log_{10}$  ratios with sign-flip tests corrected with Benjamini–Hochberg. We related site means to the transect coordinate using Spearman correlations (20 tests), and assessed agreement between predicted and MAG-derived marker abundances with Spearman correlations across 119 shotgun libraries with matched amplicon profiles (Supplementary Section S5.4).

## **Reproducibility**

The companion data descriptor covers the field, specimen, amplicon, environmental, XRF and knowledge-graph resources [26]. Inputs, scripts, machine-readable results, random seeds and checksums are maintained in the dedicated reproducibility repository at the ecology reproducibility repository. The repository links each analysis to its exact input, machine-readable result, random seed and checksum, and links shared data products to the companion resource. The shared knowledge graph, KG v3.0.1, is archived at Zenodo with its original modules, asserted export and rebuild kit. The public repository provides the staged inputs and downstream analyses, including the corrected Site 52 rainfall packages shared with the data descriptor. Its release checklist identifies the upstream amplicon, PMA, shotgun-read and genome resources and execution records required for a complete raw-read replay.

## **Software**

Amplicon reads were processed with nf-core/Ampliseq v2.14.0 (Nextflow; DADA2 and SILVA v138.2) on the KAUST Ibex cluster, where the eggNOG-mapper v2.1.12 annotation of the shotgun genome catalogue also ran. All statistical analyses were run on a Linux workstation in Python v3.11.14 with pandas v2.2.3, NumPy v2.1.3, SciPy v1.14.1, statsmodels v0.14.6, scikit-learn v1.5.2, scikit-bio v0.7.3 and Biopython v1.84; PICRUSt2 v2.4.1 predicted functional potential. Figures were drawn with Matplotlib v3.9.4 (FreeType v2.14.3). The complete environment is pinned as a conda lock file with hashed pip requirements in the reproducibility repository. Its tests check selected numerical calculations, input/output consistency and manuscript values. A complete raw-read replay requires the upstream input and execution records listed in the repository. We used Claude (Anthropic) and OpenAI Codex to assist with analysis code and manuscript editing. The authors are responsible for the analyses and text.

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## COMPETING INTERESTS

The authors declare no competing interests.

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